

Modelling Earth's Energy Budget

PhET Greenhouse Effect · PhET

Year 10

~30 minutes

Science · Earth & Space Science

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Simulation: <https://phet.colorado.edu/en/simulations/greenhouse-effect>

Curriculum links: AC9S10U04 · climate change

Learning intentions

- I can describe how greenhouse gases affect the balance between incoming sunlight and outgoing infrared radiation.
- I can use the simulation to compare Earth's energy budget under different greenhouse gas concentrations.
- I can explain, using evidence from the model, why more greenhouse gases lead to a warmer average surface temperature.

Getting started

Open the simulation's Energy Balance (or Layer Model) screen. Start with the '1750' preset (pre-industrial greenhouse gas levels) and note the surface temperature shown before you change anything.

Tasks

1. Record the surface temperature and greenhouse gas concentration shown for the '1750' preset.

2. Switch to the 'Today' preset. Record the new surface temperature and greenhouse gas concentration. What is the difference between the two temperatures?

3. Watch the infrared radiation leaving the atmosphere in both presets. Explain what changes about outgoing infrared radiation as greenhouse gas concentration increases.

4. Set the greenhouse gas concentration to three different custom levels and record the resulting surface temperature for each.

Greenhouse gas concentration (custom setting)	Resulting surface temperature

5. Based on your table, describe the relationship between greenhouse gas concentration and surface temperature.

6. Using the phrases 'incoming sunlight', 'trapped', and 'outgoing infrared radiation', explain why Earth's average temperature rises when greenhouse gas concentration increases.

7. Try setting the greenhouse gas concentration to zero (or the lowest possible setting). What happens to the surface temperature, and what does this tell you about the role greenhouse gases naturally play, even before human influence?

Extension (optional)

The simulation lets you set a custom greenhouse gas concentration higher than today's level. Use it to predict what could happen to Earth's surface temperature if concentrations continue to rise, and suggest one real-world consequence of that change.
